

Curriculum Vitae

Name: **Chin-Fei Lee**
Address: **Institute of Astronomy and Astrophysics, Academia Sinica (ASIAA)**
11F of Astronomy-Mathematics Building, NTU
No.1, Roosevelt Rd, Sec. 4
Taipei 10617, Taiwan, R.O.C.
Tel: 886-2-2366-5445
Email: cflee@asiaa.sinica.edu.tw
Webpage: <http://www.asiaa.sinica.edu.tw/people/cv.php?i=cflee>

Current Positions

2023-Present **ALMA Board Observer**
2023-Present **PI, Taiwan-US Astronomical Facility Collaboration: ALMA and GMT (80 MNTD/yr, the largest NSTC PI grant)**
2022-Present **co-PI, Taiwan Astronomical ObserVatory Alliance (TAOVA) (~ 10 MNTD/yr, NSTC)**
2022-Present **Distinguished Research Fellow, ASIAA**
2017-Present **Deputy Director, ASIAA (except for 2021/09-2022/08)**
2016-Present **Manager, ASIAA CASA Development Center for ALMA**
Here, ALMA = Atacama Large Millimeter/submillimeter Array
GMT = Giant Magellan Telescope
NSTC = National Science and Technology Council

Previous Positions

2015-2022 **Research Fellow, ASIAA**
2019-2023 **PI, ALMA-Taiwan Project (65 MNTD/yr, the largest NSTC PI grant)**
2016-2019 **co-PI, ALMA-Taiwan Project**
2015-2021 **Joint Professor, Department of Physics, National Taiwan University, Taiwan**
2009-2015 **Manager, Taiwan ARC (ALMA Regional Center) node**
2010-2012 Adjunct Associate Professor, Institute of Earth Sciences, National Taiwan Normal University, Taiwan
2010-2014 Associate Research Fellow, ASIAA
2008-2010 Adjunct Assistant Professor, Institute of Earth Sciences, National Taiwan Normal University, Taiwan
2006-2009 Assistant Research Fellow, ASIAA
2003-2006 Postdoctoral Fellow, Harvard-Smithsonian Center for Astrophysics – Submillimeter Array (SMA) Project
2001-2003 Postdoctoral Research Associate, National Research Council/NASA/Jet Propulsion Laboratory/Caltech
1995-2001 Research/Teaching Assistant, Department of Astronomy, University of Maryland
1994-1995 Research Assistant, Institute of Astronomy and Astrophysics, Academia Sinica, Taiwan
1992-1994 Research Assistant, Institute of Astronomy, National Central University, Taiwan

Education

2001	Ph. D. in Astronomy, University of Maryland, USA Dissertation Advisers: Drs. Lee Mundy and James Stone
1997	M. S. in Astronomy, University of Maryland, USA
1994	M. S. in Astronomy, National Central University, Taiwan
1992	B. S. in Physics, National Central University, Taiwan

Honors and Awards

2024	Ministry of Education's 68th Annual Academic Awards
2022	The Physical Society of Taiwan (TPS) Fellow
2021	Presidential Scholars Program, Academia Sinica (AS)
2020	Outstanding Research Award, Ministry of Science and Technology (MoST)
2019-2023	Investigator Award, Academia Sinica (AS)
2017	Outstanding Research Award, Ministry of Science and Technology (MoST)
2013-2017	Career Development Award, Academia Sinica (AS)
2001-2003	National Research Council Postdoctoral Research Associateship, NASA/JPL/Caltech
2001	Phi Kappa Phi Honor Society, University of Maryland
1995-1997	Berkeley-Illinois-Maryland Association (BIMA) scholarship from Institute of Astronomy and Astrophysics, Academia Sinica, Taiwan
1994	Phi Tau Phi Scholastic Honor Society of the Republic of China
1992-1994	Ministry of Education Graduate Fellowship, Taiwan

Services for Institute of Astronomy and Astrophysics, Academia Sinica (ASIAA)

2017-Present	Deputy Director (except for 2021)
2017-Present	Management Team (except for 2021)
2022-Present	Chair, Space Utilization Committee
2017-Present	Member, Appointment and Promotion Committee
2022-Present	Member, Performance Evaluation Committee
2022-2022	Member, Award Nomination Committee
2022-2022	Member, Mentoring Committee
2017-2022	Building and Safety Committee, Served as Chair from 2017-2021
2014-2022	Computing Committee, Served as Chair from 2014-2017
2017-2021	Performance Evaluation Committee
2016-2018	Faculty Search Committee
2015-2017	Time Allocation Committee for Canada-France-Hawaii Telescope (CFHT), ASIAA
2011-2014	Standing Committee
2007-2008	Time Allocation Committee for Submillimeter Array (SMA)

Taiwan-US Astronomical Facility Collaboration: ALMA and GMT (80 MNTD/yr)

(ALMA: Atacama Large Millimeter/submillimeter Array; GMT: Giant Magellan Telescope)

- 2019-Present **PI:** make plans for the project, propose to MoST every 3 years, and monitor the progress and budget execution. This project is consisted of ALMA and GMT components. Current major subprojects in the ALMA component are Taiwan-ARC, ACDC, Band-1 project, and GLT/VLBI/EHT project, with the first two established by myself.
- 2023-Present **ALMA Board Observer:** Attending ALMA Board Meeting and joining the discussion and planning
- 2016-Present **Manager of ASIAA CASA Development Center (ACDC):** established and manage the ACDC for software development, collaborating with NRAO in US, IDIA in South Africa, Physics Department in University of Alberta in Canada to develop the next generation visualization and analysis tool called Cube Analysis and Rendering Tool for Astronomy (CARTA), providing a fast and efficient visualization and analysis tool for ALMA and other big radio interferometry arrays, e.g., VLA, ngVLA, SKA. Since 2021, CARTA has been integrated in ALMA archive and used by users from all over the world.
- 2016-2019 **co-PI:** helped PI to plan for the project, propose to MoST every 3 years, and monitor the progress and budget execution.
- 2011-2014 **East-Asia ALMA Scientific Advisory Committee:** providing scientific advices to ALMA science operation.
- 2009-2015 **Manager of Taiwan-ALMA Regional Center (ARC):** established and managed the Taiwan-ARC node, providing user supports to in-island ALMA users and observing support to ALMA operations in Chile, helping in-island ALMA users in obtaining observing time through their proposals, providing computing resources for data reduction and analysis.

Taiwan Astronomical Observatory Alliance (TAOvA) ~ 10 MNTD/yr)

- 2022-Present **co-PI:** We aim to provide support to the community with tuCASA-like activities but extending to the entire community, including observers across the wavelengths and theoreticians, and specifically with improved access to the OIR observing facilities. We will utilize the existing telescopes in radio, such as the SMA, JCMT, and ALMA, and in OIR, such as those at Lulin Observatory, and the 2-m telescope to be ready in a couple of years to train students and to coordinate activities. There are 3 major components in this alliance: (1) a platform to serve the community at large for radio and OIR observers with facility access, training events, etc., (2) operations of Lulin Observatory to run the telescopes for astronomical observations, and (3) a financial investment to the 2-m telescope to be installed in Mexico to exchange for observing time for the entire community.

Here tuCASA: The UCAT Center for ALMA Science Advancement
where UCAT: The University Consortium of ALMA-Taiwan

Teaching Experiences

- 2015-2021 Joint Professor, Department of Physics, National Taiwan University, Taiwan
– Taught a graduate course: Radio Astronomy
- 2008-2010 Adjunct Assistant Professor, Institute of Earth Sciences, National Taiwan Normal University, Taiwan
– Taught a graduate course: Radio Astronomy
- 1995-1998 Teaching Assistant, Department of Astronomy, University of Maryland

- Led discussion sections for the following undergraduate Astronomy classes:
(1) Introduction to Astronomy, (2) Solar System Astronomy, (3) Cosmology, and (4) Astronomy and Astrophysics

Past ARC members under my supervision when I was the ARC manager (2009-2015)

Members Shigehisa Takakuwa (now Kagoshima Univ), Yu-Nung Su (now ARC manager), Alfonso Trejo-Cruz (now ARC support scientist), Baobab Hau-Yu Liu (now ASIAA faculty), Kuo-Song Wang (now ACDC support scientist), Yu-Ting Wu (ARC member), Rita Peng (outside Astronomy), Masaaki Hiramatsu (now NAOJ), Satoko Takahashi (now NAOJ posted in Chile), Ronny Zhao-Geisler (now European Space Operations Centre), Edwige Chapillon (now IRAM ARC node), Wesley Young (Retired)

System Managers Zheng Meyer-Zhao (now Cluster computing group of SURFsara in Amsterdam), Anthony Moraghan (now in SKAO at The University of Manchester)

ASIAA CASA Development Center members under my management (2016-Present)

2023/10-Present Kuan-Chou Hou (software engineer)
 2020/02-Present Yu-Hsuan Hwang (software engineer, frontend lead)
 2020/02-Present Ming-Yi Lin (software engineer)
 2019/11-Present Tian-Hao Chang (software engineer)
 2017/03-Present Cheng-Chin Chiang (software engineer)
 2016/08-Present Kuo-Song Wang (support scientist, deputy manager)
 2016/08-2024/01 Anthony Moraghan (support scientist, deputy manager) (Now in SKAO at Manchester)
 2022/06-2023/09 I-Chenn Chen (software engineer)
 2018/07-2022/02 Shou-Chieh Hsu (software engineer)
 2018/06-2021/05 Hengtai Jan (software engineer) (Now in Google)
 2017/02-2018/08 Sean Chao (software engineer) (Astronomy PhD Program)
 2017/08-2018/03 Julie Yao (software engineer)
 2016/11-2018/02 Grimmer Teng-Chieh Kang (software engineer lead)
 2017/02-2018/07 Chia-Jung Hsu (software engineer) (Astronomy PhD Program)
 2016/08-2017/04 Erh-Chung Chen (software engineer)
 2016/08-2016/10 Light Shing-Kwong Wong (software engineer)

Postdocs

2023/10-Present Shih-Ying Hsu, Star Formation
 2022/06-Present Jesús Alejandro López-Vázquez, Star Formation
 2020/04-2022/06 Po-Sheng Huang, Proplanetary Nebulae, Observations and Simulations
 2019/03-2023/06 Sommath Dutta, Star Formation
 2016-2020 Tien-Hao Hsieh, Star Formation
 2013/09-2016/08 Anthony Moraghan, Star Formation, Simulations.
 2010/08-2014/07 Alfonso Trejo-Cruz, ARC Postdoc

Students

2023-Present Jyun-Heng Lin, Undergraduate student in Physics, National Taiwan Normal University

2021-Present Ying-Chi Hu, Undergraduate and then Master's student in Physics, National Tsing-Hua University

2019-2022 Yi-Shan Tsai, Master's student in Astronomy, National Taiwan University

2014-2022 Kai-Syun Jhan, Master's student and then **PhD student** in Physics, National Taiwan University. Now in National Chung-Shan Institute of Science & Technology

2009-2019 Po-Sheng Huang, Master's student and then **PhD student** in Astronomy, National Taiwan University. Now in Korea Astronomy and Space Science Institute

2016-2018 Daniel Zhe-Yu Lin, Undergraduate student in Physics, National Chung Cheng University, (now PhD student in University of Virginia)

2016-2017 Baole Lim, Undergraduate student in Physics, National University of Singapore

2012-2016 Hsiang-Chih Hwang, Undergraduate student in Physics, National Taiwan University (PhD study at John Hopkins University, now Postdoc at Inst. of Advanced Study)

2012-2013 Jeremy Tay Jian Hao, Undergraduate student in Physics, National University of Singapore

2011-2014 Tzu-Ying Chen, Master's Student in Earth Sciences, National Taiwan Normal University

2011-2014 Hung-Jin Huang, Master's Student in Astronomy, National Taiwan University (now Postdoc in University of Arizona)

2008-2008 Ming-Yi Lin, Undergraduate Student in Earth Sciences, National Taiwan Normal University (now software engineer in ASIAA)

2008-2010 Chun-Hui Yang, Master's student in Physics, National Taiwan Normal University

2007-2011 Ming-Chien Hsu, **PhD student** in Physics, National Taiwan University. Now Associate Research Fellow in Quantum Computing center, Hona Hai Research Institute

2007-2008 Yao-Yuan Mao, Undergraduate student in Physics, National Taiwan University. Now Assistant Professor of Physics and Astronomy at the University of Utah

Journal Referee : Nature, ApJ, A&A, MNRAS, RMxAA, JKAS

Observing Experiences

- Interferometry arrays: BIMA, SMA, ALMA
- Single-dish telescopes: FCRAO, JCMT